

**SEMINAR SUMMARY:
FEBRUARY 24, 2026 - LAURA FIX
PATIENT-SPECIFIC MODELING OF CARDIOVASCULAR AND
RESPIRATORY DYNAMICS**

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1. MODELING OF CARDIOPULMONARY DYNAMICS

The utilization of a mathematical model in regard to modeling the dynamics of blood-flow through the body is informative and understood to be fairly predictive of the actual dynamics of the physiological system at hand. A standard in the field of fluid dynamics for modeling flow through a network of pipes, pumps, check valves, capacitor tanks. Ohms law in fluid dynamics follows as

$$P = Q \cdot R$$

where P is pressure head at a spot along the pipe, Q is the flow rate, and R is the resistance of the pipe. Now this is a description of the model that we were presented and is an ODE model. Generally the system can be modeled with ODEs or PDEs.

We were given two projects that were being actively worked on: prediction of dynamic cerebrovascular resistance and patient-specific modeling of respiratory mechanics in preterm infants (which seems to have new material to work on).

We then briefly went over the mechanics of orthostatics and how sudden shifts in body position accommodate for blood quickly moving throughout the system with reference to gravity such that cranial blood pressure can drop below functional levels. We then delved into the general physiology of the circulatory system as well as the autonomic nervous system, with the sympathetic and parasympathetic controls that it offers.

The model was set up as eight distinct compartments over the arterial and venous vascular systems, each of which had four complementary compartments: the head, the heart, the thorax, and systemic tissue (the remainder catchall). In a rough sketch the model was given as

$$\frac{dV}{dt} = q_{in} - q_{out}, \quad \text{where} \quad q = \frac{p_{in} - p_{out}}{R}$$

and

$$V - V_{unstr} = Cp \quad \Rightarrow \quad \frac{dp}{dt} = \frac{1}{C} \left(q_{in} - q_{out} - p \frac{dC}{dt} \right)$$

The model was then run against normal respiration followed the administration of a breathing mask giving 5% CO_2 , and plots were shown. It was shown that cranial resistance hops up after re-breathing the 5% CO_2 for some time. There seem to be future research opportunities here regarding applying the model to populations with compromised cerebrovascular reactivity. Another thought was to potentially include modern sensitivity analysis and parameter identifiability techniques.